

A. Background: Properties of Stochastic Interpolants

We note some relevant properties of stochastic interpolants for our exposition.

Boundary satisfaction. For an interpolant distribution $q_t(\mathbf{x}_t|\mathbf{x}, \epsilon)$ defined in Albergo et al. (2023), and the marginal $q_t(\mathbf{x}_t)$ as defined in Eq. (2), we can check that $q_1(\mathbf{x}_1) = p(\mathbf{x}_1)$ and $q_0(\mathbf{x}_0) = q(\mathbf{x}_0)$ so that $\mathbf{x}_1 = \epsilon$ and $\mathbf{x}_0 = \mathbf{x}$.

$$q_1(\mathbf{x}_1) = \iint q_1(\mathbf{x}_1|\mathbf{x}, \epsilon)q(\mathbf{x})p(\epsilon)d\mathbf{x}d\epsilon \quad (15)$$

$$= \iint \delta(\mathbf{x}_1 - \epsilon)q(\mathbf{x})p(\epsilon)d\mathbf{x}d\epsilon \quad (16)$$

$$= \iint q(\mathbf{x})p(\mathbf{x}_1)d\mathbf{x} \quad (17)$$

$$= p(\mathbf{x}_1) \quad (18)$$

$$q_0(\mathbf{x}_0) = \iint q_0(\mathbf{x}_0|\mathbf{x}, \epsilon)q(\mathbf{x})p(\epsilon)d\mathbf{x}d\epsilon \quad (19)$$

$$= \iint \delta(\mathbf{x}_0 - \mathbf{x})q(\mathbf{x})p(\epsilon)d\mathbf{x}d\epsilon \quad (20)$$

$$= \iint q(\mathbf{x}_0)p(\epsilon)d\mathbf{x} \quad (21)$$

$$= q(\mathbf{x}_0) \quad (22)$$

Joint distribution. The joint distribution of $\mathbf{x}$ and $\mathbf{x}_t$ is written as

$$q_t(\mathbf{x}, \mathbf{x}_t) = \int q_t(\mathbf{x}_t|\mathbf{x}, \epsilon)q(\mathbf{x})p(\epsilon)d\epsilon \quad (23)$$

Independence of joint at $t = 1$.

$$q_1(\mathbf{x}, \mathbf{x}_1) = \int q_1(\mathbf{x}_1|\mathbf{x}, \epsilon)q(\mathbf{x})p(\epsilon)d\epsilon \quad (24)$$

$$= \int \delta(\mathbf{x}_1 - \epsilon)q(\mathbf{x})p(\epsilon)d\epsilon \quad (25)$$

$$= q(\mathbf{x})p(\mathbf{x}_1) \quad (26)$$

$$= q(\mathbf{x})p(\epsilon) \quad (27)$$

in which case $\mathbf{x}_1 = \epsilon$.

B. Theorems and Derivations

B.1. Divergence Minimizer

Lemma 3. Assuming marginal-preserving interpolant and metric $D(\cdot, \cdot)$, a minimizer θ^* of Eq. (7) exists, i.e. $p_{s|t}^{\theta^*}(\mathbf{x}|\mathbf{x}_t) = q_t(\mathbf{x}|\mathbf{x}_t)$, and the minimum is 0.

Proof. We directly substitute $q_t(\mathbf{x}|\mathbf{x}_t)$ into the objective to check. First,

$$p_{s|t}^{\theta^*}(\mathbf{x}_s) = \iint q_{s|t}(\mathbf{x}_s|\mathbf{x}, \mathbf{x}_t)p_{s|t}^{\theta^*}(\mathbf{x}|\mathbf{x}_t)q_t(\mathbf{x}_t)d\mathbf{x}_td\mathbf{x} \quad (28)$$

$$= \iint q_{s|t}(\mathbf{x}_s|\mathbf{x}, \mathbf{x}_t)q_t(\mathbf{x}|\mathbf{x}_t)q_t(\mathbf{x}_t)d\mathbf{x}_td\mathbf{x} \quad (29)$$

$$\stackrel{(a)}{=} q_s(\mathbf{x}_s) \quad (30)$$

where (a) is due to definition of marginal preservation. So the objective becomes

$$\mathbb{E}_{s,t} \left[w(s,t) D(q_s(\mathbf{x}_s), p_{s|t}^{\theta^*}(\mathbf{x}_s)) \right] = \mathbb{E}_{s,t} [w(s,t) D(q_s(\mathbf{x}_s), q_s(\mathbf{x}_s))] = 0 \quad (31)$$

In general, the minimizer $q_t(\mathbf{x}|\mathbf{x}_t)$ exists. However, this does not show that the minimizer is unique. In fact, the minimizer is not unique in general because a deterministic minimizer can also exist under certain assumptions on the interpolant (see Appendix B.6). $\square$

Failure Case without Marginal Preservation. We additionally show that marginal-preservation property of the interpolant $q_{s|t}(\mathbf{x}_s|\mathbf{x}, \mathbf{x}_t)$ is important for the naïve objective in Eq. (7) to attain 0 loss (Lemma 3). Consider the failure case below where the constructed interpolant is a generalized interpolant but not necessarily marginal-preserving. Then we show that there exists a t such that $p_{s|t}^{\theta}(\mathbf{x}_s)$ can never reach $q_s(\mathbf{x}_s)$ regardless of θ .

Proposition 2 (Example Failure Case). *Let $q(\mathbf{x}) = \delta(\mathbf{x})$, $p(\epsilon) = \delta(\epsilon - 1)$, and suppose an interpolant $I_{s|t}(\mathbf{x}, \mathbf{x}_t) = (1 - \frac{s}{t})\mathbf{x} + \frac{s}{t}\mathbf{x}_t$ and $\gamma_{s|t} = \frac{s}{t}\sqrt{1 - \frac{s}{t}}\mathbb{1}(t < 1)$, then $D(q_s(\mathbf{x}_s), p_{s|t}^{\theta}(\mathbf{x}_s)) > 0$ for all $0 < s < t < 1$ regardless of the learned distribution $p_{s|t}^{\theta}(\mathbf{x}|\mathbf{x}_t)$ given any metric $D(\cdot, \cdot)$.*

Proof. This example first implies the learning target

$$q_s(\mathbf{x}_s) = \iint q_s(\mathbf{x}_s|\mathbf{x}, \epsilon) q(\mathbf{x}) p(\epsilon) d\mathbf{x} d\epsilon \quad (32)$$

$$= \iint \delta(\mathbf{x}_s - ((1-s)\mathbf{x} + s\epsilon)) \delta(\mathbf{x}) \delta(\epsilon - 1) d\mathbf{x} d\epsilon \quad (33)$$

$$= \delta(\mathbf{x}_s - s) \quad (34)$$

is a delta distribution. However, we show that if we select any $t < 1$ and $0 < s < t$, $p_{s|t}^{\theta}(\mathbf{x}_s)$ can never be a delta distribution.

$$p_{s|t}^{\theta}(\mathbf{x}_s) = \iint q_{s|t}(\mathbf{x}_s|\mathbf{x}, \mathbf{x}_t) p_{s|t}^{\theta}(\mathbf{x}|\mathbf{x}_t) q_t(\mathbf{x}_t) d\mathbf{x}_t d\mathbf{x} \quad (35)$$

$$= \iint \mathcal{N}((1 - \frac{s}{t})\mathbf{x} + \frac{s}{t}\mathbf{x}_t, \frac{s^2}{t^2}(1 - \frac{s^2}{t^2})I) p_{s|t}^{\theta}(\mathbf{x}|\mathbf{x}_t) \delta(\mathbf{x}_t - t) d\mathbf{x}_t d\mathbf{x} \quad (36)$$

$$= \int \mathcal{N}((1 - \frac{s}{t})\mathbf{x} + s, \frac{s^2}{t^2}(1 - \frac{s^2}{t^2})I) p_{s|t}^{\theta}(\mathbf{x}|\mathbf{x}_t = t) d\mathbf{x} \quad (37)$$

Now, we show the model distribution has non-zero variance under these choices of t and s . Expectations are over $p_{s|t}^{\theta}(\mathbf{x}_s)$ or conditional interpolant $q_{s|t}(\mathbf{x}_s|\mathbf{x}, \mathbf{x}_t)$ for all equations below.

$$\text{Var}(\mathbf{x}_s) = \mathbb{E}[\mathbf{x}_s^2] - \mathbb{E}[\mathbf{x}_s]^2 \quad (38)$$

$$= \iint \mathbb{E}[\mathbf{x}_s^2|\mathbf{x}, \mathbf{x}_t] p_{s|t}^{\theta}(\mathbf{x}|\mathbf{x}_t) q_t(\mathbf{x}_t) d\mathbf{x} d\mathbf{x}_t - \mathbb{E}[\mathbf{x}_s]^2 \quad (39)$$

$$= \iint \mathbb{E}[\mathbf{x}_s^2|\mathbf{x}, \mathbf{x}_t] p_{s|t}^{\theta}(\mathbf{x}|\mathbf{x}_t) q_t(\mathbf{x}_t) d\mathbf{x} d\mathbf{x}_t - \mathbb{E}[\mathbf{x}_s]^2 \quad (40)$$

$$= \iint [\text{Var}(\mathbf{x}_s|\mathbf{x}, \mathbf{x}_t) + \mathbb{E}[\mathbf{x}_s|\mathbf{x}, \mathbf{x}_t]^2] p_{s|t}^{\theta}(\mathbf{x}|\mathbf{x}_t) q_t(\mathbf{x}_t) d\mathbf{x} d\mathbf{x}_t - 2\mathbb{E}[\mathbf{x}_s]^2 + \mathbb{E}[\mathbf{x}_s]^2 \quad (41)$$

$$= \iint [\text{Var}(\mathbf{x}_s|\mathbf{x}, \mathbf{x}_t) + \mathbb{E}[\mathbf{x}_s|\mathbf{x}, \mathbf{x}_t]^2] p_{s|t}^{\theta}(\mathbf{x}|\mathbf{x}_t) q_t(\mathbf{x}_t) d\mathbf{x} d\mathbf{x}_t \quad (42)$$

$$- \iint \mathbb{E}[\mathbf{x}_s] \mathbb{E}[\mathbf{x}_s|\mathbf{x}, \mathbf{x}_t] p_{s|t}^{\theta}(\mathbf{x}|\mathbf{x}_t) q_t(\mathbf{x}_t) d\mathbf{x} d\mathbf{x}_t + \mathbb{E}[\mathbf{x}_s]^2$$

$$= \iint \underbrace{[\text{Var}(\mathbf{x}_s|\mathbf{x}, \mathbf{x}_t) + \mathbb{E}[\mathbf{x}_s|\mathbf{x}, \mathbf{x}_t]^2 - \mathbb{E}[\mathbf{x}_s] \mathbb{E}[\mathbf{x}_s|\mathbf{x}, \mathbf{x}_t] + \mathbb{E}[\mathbf{x}_s]^2]}_{(a)} p_{s|t}^{\theta}(\mathbf{x}|\mathbf{x}_t) q_t(\mathbf{x}_t) d\mathbf{x} d\mathbf{x}_t \quad (43)$$

where (a) can be simplified as

$$\text{Var}(\mathbf{x}_s|\mathbf{x}, \mathbf{x}_t) + \left(\mathbb{E}[\mathbf{x}_s|\mathbf{x}, \mathbf{x}_t]^2 - \mathbb{E}[\mathbf{x}_s] \right)^2 > 0 \quad (44)$$

because $\text{Var}(\mathbf{x}_s|\mathbf{x}, \mathbf{x}_t) > 0$ for all $0 < s < t < 1$ due to its non-zero Gaussian noise. Therefore, $\text{Var}(\mathbf{x}_s) > 0$, implying $p_{s|t}^\theta(\mathbf{x}_s)$ can never be a delta function regardless of model $p_{s|t}^\theta(\mathbf{x}|\mathbf{x}_t)$. A valid metric $D(\cdot, \cdot)$ over probability $p_{s|t}^\theta(\mathbf{x}_s)$ and $q_s(\mathbf{x}_s)$ implies

$$D(q_s(\mathbf{x}_s), p_{s|t}^\theta(\mathbf{x}_s)) = 0 \iff p_{s|t}^\theta(\mathbf{x}_s) = q_s(\mathbf{x}_s)$$

which means

$$p_{s|t}^\theta(\mathbf{x}_s) \neq q_s(\mathbf{x}_s) \implies D(q_s(\mathbf{x}_s), p_{s|t}^\theta(\mathbf{x}_s)) > 0$$

□

B.2. Boundary Satisfaction of Model Distribution

The operator output $p_{s|t}^\theta(\mathbf{x}_s)$ satisfies boundary condition.

Lemma 4 (Boundary Condition). *For all $s \in [0, 1]$ and all θ , the following boundary condition holds.*

$$q_s(\mathbf{x}_s) = p_{s|s}^\theta(\mathbf{x}_s) \quad (45)$$

Proof.

$$\begin{aligned} p_{s|s}^\theta(\mathbf{x}_s) &= \iint \underbrace{q_{s,s}(\mathbf{x}_s|\mathbf{x}, \bar{\mathbf{x}}_s)}_{\delta(\mathbf{x}_s - \bar{\mathbf{x}}_s)} p_{s|s}^\theta(\mathbf{x}|\bar{\mathbf{x}}_s) q_{s,1}(\bar{\mathbf{x}}_s) d\bar{\mathbf{x}}_s d\mathbf{x} \\ &= \int p_{s|s}^\theta(\mathbf{x}|\mathbf{x}_s) q_s(\mathbf{x}_s) d\mathbf{x} \\ &= q_s(\mathbf{x}_s) \int p_{s|s}^\theta(\mathbf{x}|\mathbf{x}_s) d\mathbf{x} \\ &= q_s(\mathbf{x}_s) \end{aligned}$$

□

B.3. Definition of Well-Conditioned $r(s, t)$

For simplicity, the mapping function $r(s, t)$ is well-conditioned if

$$r(s, t) = \max(s, t - \Delta(t)) \quad (46)$$

where $\Delta(t) \geq \epsilon > 0$ is a positive function such that $r(s, t)$ is increasing for $t \geq c_0(s)$ where $c_0(s)$ is the largest t that is mapped to s . Formally, $c_0(s) = \sup\{t : r(s, t) = s\}$. For $t \geq c_0(s)$, the inverse w.r.t. t exists, i.e. $r^{-1}(s, \cdot)$ and $r^{-1}(s, r(s, t)) = t$. All practical implementations follow this general form, and are detailed in Appendix C.6.

B.4. Main Theorem

Theorem 1. *Assuming $r(s, t)$ is well-conditioned, the interpolant is marginal-preserving, and θ_n^* is a minimizer of Eq. (8) for each n with infinite data and network capacity, for all $t \in [0, 1]$, $s \in [0, t]$,*

$$\lim_{n \rightarrow \infty} \text{MMD}^2(q_s(\mathbf{x}_s), p_{s|t}^{\theta_n^*}(\mathbf{x}_s)) = 0. \quad (9)$$

Proof. We prove by induction on sequence number n . First, $r(s, t)$ is well-conditioned by following the definition in Eq. (46). Furthermore, for notational convenience, we let $r_n^{-1}(s, \cdot) := r^{-1}(s, r^{-1}(s, r^{-1}(s, \dots)))$ be n nested application of $r^{-1}(s, \cdot)$ on the second argument. Additionally, $r_0^{-1}(s, t) = t$.